ALGEBRA 2 · CONIC SECTIONS

Conic Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Conic Explorer** on Mr. K's Classroom: Algebra 2 → Conic Sections → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. What do you think happens to an ellipse when a = b?

2 Explore use the tool and record what you see

Identify and describe

Build each conic in the tool.

Equation	Conic	Center or vertex	Key facts
$(x - 2)^2 + (y + 1)^2 = 25$			
$\frac{x^2}{25} + \frac{y^2}{9} = 1$			
$\frac{x^2}{9} + \frac{y^2}{25} = 1$			
$\frac{x^2}{4} - \frac{y^2}{16} = 1$			
$y^2 = -8x$			
$(x - 1)^2 = 4(y + 3)$			

3 Write the rule fill in each blank from what you observed

1. A circle with center (h, k) and radius r: $(x - h)^2 + (y - k)^2 =$ _____.
2. For an ellipse, $c^2 =$ _____. For a hyperbola, $c^2 =$ _____.
3. In an ellipse, the larger denominator tells the direction of the _____ axis.
4. The asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ have slopes _____.
5. Every point on a parabola is the same distance from the _____ and the _____.

4 Check no tool: show your work

1. Center and radius of $x^2 + (y - 3)^2 = 49$?

2. Foci of $\frac{x^2}{100} + \frac{y^2}{36} = 1$?

3. Asymptotes of $\frac{y^2}{9} - \frac{x^2}{4} = 1$?

4. Focus of $x^2 = 12y$?

5 Reflect

How can you tell an ellipse from a hyperbola by looking at the equation?